Rates and percents

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 5.2 × 10

2. 500 ÷ 100

3. Double 5.5

4. Half of 30

5. A walker travels 12 miles in 3 hours. Find miles per hour.

6. 9 of 12 tiles are green. What percent are green?

7. The pairs (2, 7), (4, 14), and (6, __) have the same ratio. Complete the table.

8. A line marks 0, 3, and 6. How far from 3 toward 6 is $4\frac{1}{2}$?

9. $\frac{1}{4} + \frac{1}{4}$

10. 0.7 × 100

11. 8 × 3

12. A box is 8 cm by 3 cm by 3 cm. Find its volume.

13. A mystery number has this property: 25% of it is 6 more than 10% of it. Find the number.

14. A cyclist travels 36 kilometers in 6 hours at steady speed. Find kilometers per hour.

15. 6 markers cost $3.3. Find dollars per marker.

16. A printer makes 20 cards in 4 minutes. Find cards per minute and minutes per card.

17. A runner goes 3 miles in 15 minutes at steady speed. Find miles per hour.

18. Pack A: 6 pencils for $3.3. Pack B: 8 for $4.8. Which costs less per pencil?

19. A robot rolls 3 meters per second. Complete distances for 0, 2, and 4 seconds.

20. Find 20% of 40. Name the whole.

21. 20 stickers are 25% of a collection. How many stickers are in the whole collection?

22. 28 of 80 books are mysteries. What percent are mysteries?

23. Find 25% of $16. Give a money answer.

CHALLENGE / Explain your plan and check the result

C1. Flowers occupy 40% of a garden and cover 24 square meters. Beans use $\frac{2}{3}$ of the remaining area. Find the whole area, bean area, and area left for paths.

C2. Cyclist A goes 10 km in 40 minutes; B goes 9 km in 30 minutes. Find both hourly rates. How long would B need for 12 km?